

Determinants of Matrices

Algebra Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Find the determinant of a 2×2 matrix using the formula $ad - bc$
- Set up and evaluate the determinant of a 3×3 matrix using the diagonal method
- Apply determinant concepts including matrices with negative and zero entries

Problems

1. Find the determinant of matrix A.

$$\begin{vmatrix} 3 & 1 \\ 2 & 4 \end{vmatrix}$$

2. Find the determinant of matrix B.

$$\begin{vmatrix} 5 & 2 \\ 3 & 1 \end{vmatrix}$$

3. Find the determinant of matrix C.

$$\begin{vmatrix} 6 & 0 \\ 0 & 6 \end{vmatrix}$$

4. Find the determinant of matrix D, which contains negative values.

$$\begin{vmatrix} -2 & 3 \\ 4 & -1 \end{vmatrix}$$

5. Find the determinant of matrix E.



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Matrices — Determinants Worksheet

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$$\begin{vmatrix} 7 & -3 \\ 2 & 5 \end{vmatrix}$$

6. Find the determinant of matrix F. Notice what is special about this matrix and its determinant.

$$\begin{vmatrix} 4 & 8 \\ 1 & 2 \end{vmatrix}$$

7. Find the determinant of the 3×3 matrix G using the diagonal method. Copy the first two columns to the right of the matrix, find the three main diagonal products and add them, then subtract the three anti-diagonal products.

$$\begin{vmatrix} 1 & 2 & 0 \\ 3 & 1 & 2 \\ 0 & 1 & 3 \end{vmatrix}$$

7. Find the determinant of the 3×3 matrix G using the diagonal (Sarrus) method.

$$\begin{vmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{vmatrix}$$

8. Find the determinant of the 3×3 matrix H using the diagonal method.

$$\begin{vmatrix} 3 & 1 & 2 \\ 0 & 2 & 1 \\ 1 & 0 & 3 \end{vmatrix}$$

9. Find the determinant of the 3×3 matrix J, which contains negative values.



$$\begin{vmatrix} 2 & -1 & 3 \\ 1 & 4 & -2 \\ -3 & 0 & 1 \end{vmatrix}$$

10. Given that the determinant of matrix K equals $2x$ minus 6, and the matrix below has a determinant that must equal 10, find the value of x .

$$\det(K) = 2x - 6 = 10$$



Determinants of Matrices — Answer Key

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Answer Key

1. Answer: 10

- Use the formula: $\det(A) = ad - bc$
- $\det(A) = (3)(4) - (1)(2) = 12 - 2 = 10$

2. Answer: -1

- Use the formula: $\det(B) = ad - bc$
- $\det(B) = (5)(1) - (2)(3) = 5 - 6 = -1$

3. Answer: 36

- Use the formula: $\det(C) = ad - bc$
- $\det(C) = (6)(6) - (0)(0) = 36 - 0 = 36$

4. Answer: -10

- Use the formula: $\det(D) = ad - bc$
- $\det(D) = (-2)(-1) - (3)(4) = 2 - 12 = -10$

5. Answer: 41

- Use the formula: $\det(E) = ad - bc$
- $\det(E) = (7)(5) - (-3)(2) = 35 + 6 = 41$

6. Answer: 0

- Use the formula: $\det(F) = ad - bc$
- $\det(F) = (4)(2) - (8)(1) = 8 - 8 = 0$
- A determinant of 0 means the matrix is singular (non-invertible)

7. Answer: 7

- Copy columns 1 and 2 to the right of the matrix
- Main diagonals: $(1)(1)(3) + (2)(2)(0) + (0)(3)(1) = 3 + 0 + 0 = 3$
- Anti-diagonals: $(0)(1)(0) + (1)(2)(3) + (3)(3)(2) = 0 + 6 + \dots$ wait, recalculate carefully
- Main: $(1 \cdot 1 \cdot 3) + (2 \cdot 2 \cdot 0) + (0 \cdot 3 \cdot 1) = 3 + 0 + 0 = 3$
- Anti: $(0 \cdot 1 \cdot 0) + (1 \cdot 2 \cdot 3) + (3 \cdot 3 \cdot 2) = 0 + 6 + \dots$ re-examine: anti-diagonals go bottom-left to top-right
- Anti: $(0)(1)(1) + (3)(2)(0) + (1)(3)(2) = 0 + 0 + 6 = \dots$ let's use cofactor check: $\det = 1(3-2) - 2(9-0) + 0 = 1 - 18 + 0$
- ...
- Using cofactor expansion along row 1: $1 \cdot \det([[1,2],[1,3]]) - 2 \cdot \det([[3,2],[0,3]]) + 0$
- $= 1 \cdot (3-2) - 2 \cdot (9-0) + 0 = 1 - 18 = -17$... re-check original entries
- Diagonal method: Main = $1 \cdot 1 \cdot 3 + 2 \cdot 2 \cdot 0 + 0 \cdot 3 \cdot 1 = 3$; Anti = $0 \cdot 1 \cdot 0 + (\text{bottom-left diag}) 1 \cdot 2 \cdot 3 + 3 \cdot 3 \cdot 2$ — wait re-read diagonals
- Correct diagonal method: Main sums = $(1 \cdot 1 \cdot 3) + (2 \cdot 2 \cdot 0) + (0 \cdot 3 \cdot 1) = 3 + 0 + 0 = 3$; Anti sums = $(0 \cdot 1 \cdot 0) + (1 \cdot 2 \cdot 3) + (3 \cdot 3 \cdot 2) \dots$
- Anti = $(0)(1)(3\text{rd col top})$ — using standard rule: anti =



$(\text{row3col1} \cdot \text{row2col2} \cdot \text{row1col3}) + (\text{row3col2} \cdot \text{row2col3} \cdot \text{row1col1}) \dots$ not matching; use cofactor: $\det = 1(1 \cdot 3 - 2 \cdot 1) - 2(3 \cdot 3 - 2 \cdot 0) + 0 = 1(1) - 2(9) = 1 - 18 = -17$; answer is -17

7. Answer: 9

- Copy columns 1 and 2 to the right of the matrix to set up the diagonal method
- Main diagonal products: $(2)(3)(2) + (1)(1)(0) + (0)(1)(1) = 12 + 0 + 0 = 12$
- Anti-diagonal products: $(0)(3)(0) + (1)(1)(2) + (2)(1)(1) = 0 + 2 + 2 = \dots$ re-examine
- Anti (bottom-left to top-right): $(0)(3)(0) + (1)(1)(2) + (2)(1)(1)$
- $= 0 + 2 + 2 = 4 \dots$ recalculate: $\text{anti} = \text{row3c1} \cdot \text{row2c2} \cdot \text{row1c3} + \text{row3c2} \cdot \text{row2c3} \cdot \text{row1c1} + \text{row3c3} \cdot \text{row2c1} \cdot \text{row1c2}$
- $= (0)(3)(0) + (1)(1)(2) + (2)(1)(1) = 0 + 2 + 2 = 4 \dots$ still 4
- $\det = 12 - 4 = 8 \dots$ check with cofactor: $2(6-1) - 1(2-0) + 0 = 10 - 2 = 8$; answer is 8

8. Answer: 13

- Copy columns 1 and 2 to the right: extended array is $[3, 1, 2, 3, 1 / 0, 2, 1, 0, 2 / 1, 0, 3, 1, 0]$
- Main diagonals (top-left to bottom-right): $(3)(2)(3) + (1)(1)(1) + (2)(0)(0) = 18 + 1 + 0 = 19$
- Anti-diagonals (top-right to bottom-left): $(2)(2)(1) + (3)(1)(0) + (1)(0)(3) \dots$ use (col3, col2, col1) of anti: $(2)(2)(1) + (1)(0) \dots$
- Anti: $\text{row1c3} \cdot \text{row2c2} \cdot \text{row3c1} + \text{row1c1} \cdot \text{row2c3} \cdot \text{row3c2} + \text{row1c2} \cdot \text{row2c1} \cdot \text{row3c3} = (2)(2)(1) + (3)(1)(0) + (1)(0)(3) = 4 + 0 + 0 = 4 \dots$ hmm
- Check with cofactor expansion: $3(2 \cdot 3 - 1 \cdot 0) - 1(0 \cdot 3 - 1 \cdot 1) + 2(0 \cdot 0 - 2 \cdot 1) = 3(6) - 1(-1) + 2(-2) = 18 + 1 - 4 = 15$
- Re-check anti: $\text{anti} = (2)(2)(1) + (3)(1)(0) + (1)(0)(3) \dots$ standard Sarrus anti $= (2 \cdot 2 \cdot 1) + (1 \cdot 1 \cdot \dots$
- Sarrus anti-diags starting from top-right going down-left: $r1c3 \cdot r2c2 \cdot r3c1 = 2 \cdot 2 \cdot 1 = 4$; $r1c1 \cdot r2c3 \cdot r3c2 = 3 \cdot 1 \cdot 0 = 0$; $r1c2 \cdot r2c1 \cdot r3c3 = 1 \cdot 0 \cdot 3 = 0$; total anti = 4
- $\det = 19 - 4 = 15$; answer is 15

9. Answer: 47

- Use the diagonal (Sarrus) method or cofactor expansion
- Main diagonals: $(2)(4)(1) + (-1)(-2)(-3) + (3)(1)(0) = 8 + (-6) + 0 = 2$
- Anti-diagonals: $(3)(4)(-3) + (2)(-2)(0) + (-1)(1)(1) = -36 + 0 + (-1) = -37$
- $\det = 2 - (-37) = 2 + 37 = 39 \dots$ verify with cofactor: $2(4 \cdot 1 - (-2) \cdot 0) - (-1)(1 \cdot 1 - (-2)(-3)) + 3(1 \cdot 0 - 4 \cdot (-3))$
- $= 2(4) + 1(1-6) + 3(0+12) = 8 + (-5) + 36 = 39$; answer is 39

10. Answer: x = 8

- Set the determinant expression equal to 10: $2x - 6 = 10$
- Add 6 to both sides: $2x = 16$
- Divide both sides by 2: $x = 8$

